

QA Engineer in 30 Days

Manual testing → API testing with Postman → Selenium automation → CI/CD pipeline.
Each day: practical tasks + a short YouTube video (≤ 25 min) to watch and follow.

- 30 DaysFull roadmap
- 4 WeeksClear phases
- ManualWeeks 1–2
- AutomationWeeks 3–4
- CI/CDAuto on push

Windows Setup for QA Testing

Tools for manual and automated testing

1	Install Java 21 LTS adoptium.net → Windows x64 MSI → required for Selenium tests adoptium.net	5	Create Maven Project IntelliJ → New Project → Maven → add selenium-java + testng to pom.xml
2	Install IntelliJ IDEA jetbrains.com/idea → Community Edition → free IDE for automation jetbrains.com	6	Install JIRA (free) atlassian.com/jira → free tier → use for test case management + bug tracking atlassian.com
3	Install Postman postman.com/downloads → API testing tool — create free account postman.com	7	Practice Site demo.opencart.com or saucedemo.com — public demo sites built for testing saucedemo.com
4	Install Chrome & ChromeDriver google.com/chrome → install Chrome (WebDriverManager auto-downloads driver) google.com/chrome	8	Key Maven pom.xml deps selenium-java 4.x · testng · extentreports · rest-assured · poi (Excel)

WEEK 1 —Manual Testing Foundations Days 1–7

Day 1Manual What is Software Testing? - Testing vs debugging — key difference - Verification vs Validation explained - Types: functional, non-functional, regression  ISTQB · Software Testing Intro · ~15min	Day 2Manual SDLC & Testing Role - SDLC phases: plan, design, build, test, deploy - Where QA fits in each phase - V-Model: each dev phase has a test phase  Software Testing Mentor · SDLC · ~20min	Day 3Manual Test Cases & Test Plans - Test case structure: ID, steps, expected result - Test plan: scope, approach, resources, schedule - Write 5 test cases for a login page  SDET · Test Cases · ~18min
Day 4Manual Test Techniques - Equivalence partitioning: valid & invalid classes - Boundary value analysis: edges matter most - Decision table for complex conditions  SDET · Test Techniques · ~20min	Day 5Manual Bug Reporting - Bug lifecycle: new → open → fixed → closed - Bug report fields: title, steps, expected, actual - Write 3 detailed bug reports from a demo app  Naveen AutomationLabs · Bug Report · ~15mi..	Day 6Manual Types of Testing - Smoke test: basic functionality works? - Regression test: did new code break old features? - UAT: does it meet user requirements?  Software Testing Mentor · Test Types · ~18..
Day 7Project Week 1 Project: Test a Live Website - Pick any e-commerce site (e.g. demo.opencart.com) - Write 10 test cases: login, cart, checkout - Log 3 real bugs found — with full bug reports  SDET · Manual Testing Project · ~20min		

WEEK 2 —API Testing, SQL & Agile QA Days 8–14

Day 8API API Testing with Postman - Install Postman — create a workspace - Send GET, POST, PUT, DELETE requests - Understand request body, headers, params  Traversy · Postman Crash · ~20min	Day 9API Postman Assertions & Collections - Write pm.test() assertions on status code - Save requests in a Collection — run all at once - Generate API test report from Collection Runner  Naveen AutomationLabs · Postman · ~20min	Day 10DB SQL for Testers - SELECT, WHERE, ORDER BY basics - JOIN: verify data across related tables - INSERT and DELETE to set up / tear down test data  Programming with Mosh · SQL · ~20min
Day 11Tools Test Management with JIRA - Create test project in JIRA (free tier) - Write test cases as JIRA tasks or Zephyr tickets - Log defects: link to test case, set priority  SDET · JIRA for QA · ~18min	Day 12Agile Agile Testing & Scrum - Sprint testing: test what devs built this sprint - Daily standup: blockers and test status - Definition of Done: all test cases passed  Software Testing Mentor · Agile QA · ~18mi..	Day 13Perf Performance Testing Concepts - Load test: normal expected user load - Stress test: push beyond limits - Use k6 or JMeter for basic load test demo  Traversy · k6 Load Test · ~15min
Day 14Project Week 2 Project: Full Manual Cycle - Use a REST API (reqres.in) — full Postman suite 20 API test cases with assertions — run all - Log results in a test report spreadsheet  Naveen AutomationLabs · API Project · ~20m..		

WEEK 3 — Selenium Automation Setup			Days 15–21
Day 15 Auto Intro to Test Automation - When to automate vs keep manual - Pyramid: unit > integration > UI tests - Install Java 21 + IntelliJ + Maven for Selenium SDET · Automation Intro · ~20min	Day 16 Selenium Selenium WebDriver Setup - Add selenium-java + WebDriverManager to pom.xml - Launch Chrome browser in code — new ChromeDriver() - Navigate to URL and close browser Naveen AutomationLabs · Selenium Setup · ~20m..	Day 17 Selenium Locators: Finding Elements - id, name, className, tagName locators - cssSelector — most reliable for complex elements - xpath — navigate parent/child relationships SDET · Locators Guide · ~20min	
Day 18 Selenium WebDriver Actions - driver.findElement().click() and .sendKeys() - driver.findElement().getText() — read page content - Dropdown: Select class — selectByVisibleText Naveen AutomationLabs · WebDriver · ~20min	Day 19 Selenium Waits: Sync with the Browser - Implicit wait: wait up to N seconds globally - Explicit wait: WebDriverWait + ExpectedConditions - Avoid Thread.sleep() — use proper waits only SDET · Selenium Waits · ~18min	Day 20 TestNG TestNG Framework - Add TestNG dependency to pom.xml @Test, @BeforeMethod, @AfterMethod annotations - Run tests — see pass/fail in TestNG report Naveen AutomationLabs · TestNG · ~20min	
Day 21 Project Week 3 Project: Automate Login - Automate login test on demo site - Assert: URL changes after successful login - Assert: error message shown for wrong password SDET · First Selenium Test · ~20min			
WEEK 4 — Framework, Reports & CI/CD			Days 22–30
Day 22 Framework Page Object Model (POM) - Separate page logic from test logic - LoginPage class: locators + action methods - Test class only calls page methods — stays clean Naveen AutomationLabs · POM · ~22min	Day 23 Framework Data-Driven Testing @DataProvider in TestNG — supply test data - Read test data from Excel with Apache POI - Run same test with 5 user accounts in one run SDET · Data Driven · ~20min	Day 24 Reporting Test Reporting with Extent - Add ExtentReports dependency to pom.xml - Log pass/fail with screenshots on failure - Open HTML report after test run in browser Naveen AutomationLabs · Reports · ~20min	
Day 25 Selenium Cross-Browser Testing - Parametrize browser: Chrome, Firefox, Edge - WebDriverManager handles all driver downloads - Run the same suite on 3 browsers in one command SDET · Cross Browser · ~15min	Day 26 API Auto API Automation with RestAssured - Add rest-assured dependency to pom.xml - given().when().get(url).then().statusCode(200) - Extract JSON response body and assert field values Naveen AutomationLabs · RestAssured · ~20m..	Day 27 CI/CD CI with GitHub Actions - Push project to GitHub repository - Add workflow: checkout → setup-java → mvn test - Tests run automatically on every push SDET · CI for QA · ~15min	
Day 28 Mobile Mobile Testing Concepts - Native vs hybrid vs web mobile apps - Appium: extends WebDriver for mobile - Manual test checklist for mobile — gestures, orientation Naveen AutomationLabs · Appium Intro · ~20..	Day 29 Security Security Testing Basics - OWASP Top 10: know the list and meanings - SQL injection test: enter ' OR 1=1 -- in inputs - XSS test: enter <script>alert(1)</script> SDET · Security Testing · ~15min	Day 30 Project Final Project: Full QA Suite 10 manual test cases + 10 automated tests - POM structure, data-driven, Extent HTML report - Push to GitHub — CI runs on every commit Naveen AutomationLabs · QA Final · ~25min	
Completion goal: Deliver a complete QA suite: manual test cases, automated Selenium tests with POM, HTML report, and GitHub CI running on every push.			